

1. The field  $\mathbf{E}(t) = 5 \sin(20 \times 10^9 t + kz) \hat{\mathbf{y}}$  (V/m) describes a wave propagating in a medium which has the dielectric constant  $\epsilon_r = 3$ . Answer the following questions concerning this wave. Be sure to include dimensions in your answers:

(a) What is the frequency  $f$  of the wave, (b) the wave amplitude, (c) the propagation direction, (d) the phase velocity of the wave  $v_p$ , (e) the wavelength  $\lambda$  of the wave, (f) the intrinsic impedance of the wave  $\eta$ , (g) the wave number (propagation constant)  $k$ ?

2. In lossy insulating material a 2.45 GHz plane wave is propagating in a casserole in the  $z$ -direction and polarized in the  $y$ -direction, with an amplitude of 20 V/m, a conductivity  $\sigma = 2.0$  S/m, and a dielectric constant  $\epsilon_r = 4$ . Find the loss tangent. Hint: this is a trivial question but the loss tangent is of primary importance when ordering a material.

3. The first step in analytically solving for the fields in any situation in electromagnetics is to obtain the vector Helmholtz equation from Maxwell's equations. In class, starting with the Maxwell's equations the electric field vector Helmholtz equation was derived. Starting with Maxwell's equations derive the vector Helmholtz equation for the magnetic field. Write down and number Maxwell's equations and at each step in the derivation tell which equation you are going to use for the next step. If you use a vector identity, before using it write down the identity from those in the back cover of the book. Be sure to use the proper vector symbol and a line above the letter to indicate each vector. You had this problem on the last homework but go ahead and do it again.

4. In each case below (i) show mathematically that the field given is a straight line, a circle or an ellipse (ii) find an expression for the angle  $\psi$  between the field vector and the axis. (iii) In each case describe the type of polarization. Hint: Remember the polarization is found by finding the phase of one of field components with respect to the other. A trick to find the difference in phase is to add or subtract a phase to both field components. By doing this you can cause the first component to have zero phase and the other component will have the difference in the two phases.

(a)  $\mathbf{E} = (j\hat{\mathbf{x}} + \hat{\mathbf{y}})e^{-jkz}$

(b)  $\mathbf{E} = [(1 + j)\hat{\mathbf{y}} + (1 - j)\hat{\mathbf{z}}]e^{-jkx}$

(c)  $\mathbf{E} = [(2 + j)\hat{\mathbf{x}} + (3 - j)\hat{\mathbf{z}}]e^{-jky}$

(d)  $\mathbf{E} = (j\hat{\mathbf{x}} + j2\hat{\mathbf{y}})e^{-jkz}$

Hint: The  $\psi = 0$  axis is determined by the right hand rule. If the wave travels in the  $z$  direction then  $x \rightarrow y \rightarrow z$  says  $\psi = 0$  is the  $x$  axis. But if the wave travels in the  $y$  direction then  $z \rightarrow x \rightarrow y$  says  $\psi = 0$  is the  $z$  axis.